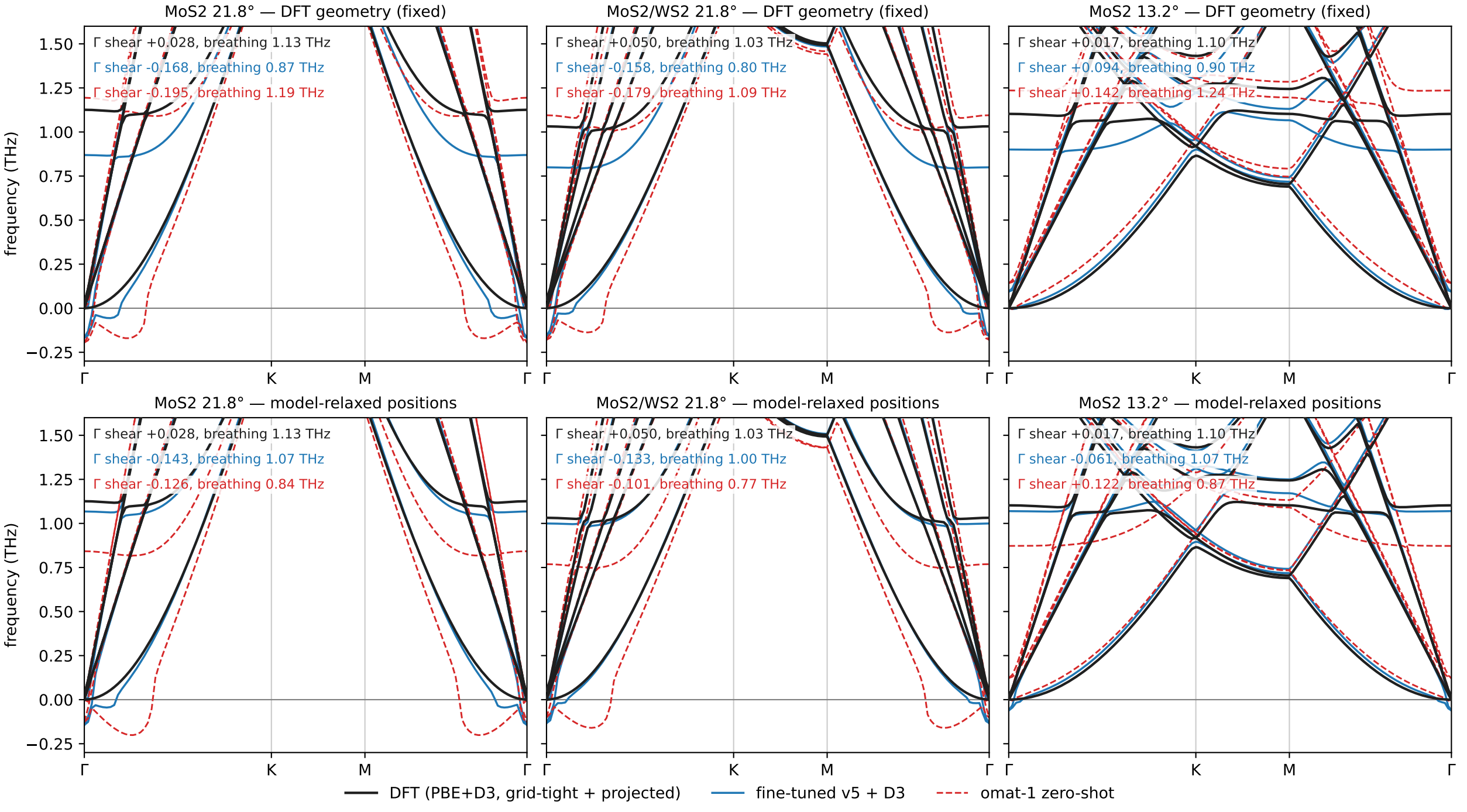

Twisted bilayers, low-frequency window: the fine-tuned model's Γ shear pair is imaginary (negative) where DFT is small and positive
MoS2 21.8° — DFT geometry (fixed)
MoS2/WS2 21.8° — DFT geometry (fixed)
MoS2 13.2° — DFT geometry (fixed)
Γ shear +0.028, breathing 1.13 THz
Γ shear -0.168, breathing 0.87 THz
Γ shear -0.195, breathing 1.19 THz
Γ shear +0.050, breathing 1.03 THz
Γ shear -0.158, breathing 0.80 THz
Γ shear -0.179, breathing 1.09 THz
Γ shear +0.017, breathing 1.10 THz
Γ shear +0.094, breathing 0.90 THz
Γ shear +0.142, breathing 1.24 THz
MoS2 21.8° — model-relaxed positions
MoS2/WS2 21.8° — model-relaxed positions
MoS2 13.2° — model-relaxed positions
Γ shear +0.028, breathing 1.13 THz
Γ shear -0.143, breathing 1.07 THz
Γ shear -0.126, breathing 0.84 THz
Γ shear +0.050, breathing 1.03 THz
Γ shear -0.133, breathing 1.00 THz
Γ shear -0.101, breathing 0.77 THz
Γ shear +0.017, breathing 1.10 THz
Γ shear -0.061, breathing 1.07 THz
Γ shear +0.122, breathing 0.87 THz
frequency (THz)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
Γ
K
M
Γ
DFT (PBE+D3, grid-tight + projected)
fine-tuned v5 + D3
omat-1 zero-shot